

INTERNOVA DATA ANALYSIS INTERNSHIP

Week 3 Assignment Report — SQL & Excel for Data Analytics

Student: Sai Chowdary **Total:** 100 Marks

This report includes evidence images generated from the SQL datasets and Excel workbooks prepared for the Week 3 tasks. The images are included so the report is complete even if you did not capture many screenshots yourself.

Task 1 — Introduction to Databases & SELECT — 10 Marks

Created a Students table and used SELECT statements and aliases.

Task 1 — SELECT

SQL: SELECT * FROM Students;

Student_ID	Name	Age	Branch	Marks
101	Sai	19	CSE	85
102	Rahul	20	ECE	78
103	Priya	19	CSE	92
104	Arjun	21	IT	88
105	Anjali	20	CSE	75
106	Kiran	19	EEE	81
107	Sneha	20	CSE	90
108	Ravi	21	IT	72
... 2 more row(s)				

SQL: SELECT Name, Branch, Marks AS Score FROM Students;

Name	Branch	Score
Sai	CSE	85
Rahul	ECE	78
Priya	CSE	92
Arjun	IT	88
Anjali	CSE	75
Kiran	EEE	81
Sneha	CSE	90
Ravi	IT	72
... 2 more row(s)		

Task 2 — WHERE, ORDER BY & Aggregate Functions — 15 Marks

Used WHERE, comparison operators, ORDER BY, COUNT, SUM, AVG, MIN and MAX.

Task 2 — WHERE / ORDER BY / Aggregates

SQL: SELECT * FROM Students WHERE Marks > 80;

Student_ID	Name	Age	Branch	Marks
101	Sai	19	CSE	85
103	Priya	19	CSE	92
104	Arjun	21	IT	88
106	Kiran	19	EEE	81
107	Sneha	20	CSE	90
109	Meena	19	ECE	86
110	Vijay	20	CSE	95

SQL: SELECT * FROM Students ORDER BY Marks DESC;

Student_ID	Name	Age	Branch	Marks
110	Vijay	20	CSE	95
103	Priya	19	CSE	92
107	Sneha	20	CSE	90
104	Arjun	21	IT	88
109	Meena	19	ECE	86
101	Sai	19	CSE	85
106	Kiran	19	EEE	81
102	Rahul	20	ECE	78
... 2 more row(s)				

SQL: SELECT COUNT(*), AVG(Marks), MIN(Marks), MAX(Marks) FROM Students;

COUNT(*)	AVG(Marks)	MIN(Marks)	MAX(Marks)
10	84.2	72	95

Task 3 — GROUP BY & HAVING — 10 Marks

Grouped students by branch and filtered grouped results using HAVING.

Task 3 — GROUP BY / HAVING

SQL: SELECT Branch, COUNT(*) AS Total_Students, AVG(Marks) AS Average_Marks FROM Students GROUP BY Branch;

Branch	Total_Students	Average_Marks
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CSE	5	87.4
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ECE	2	82.0
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EEE	1	81.0
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IT	2	80.0
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SQL: SELECT Branch, AVG(Marks) AS Average_Marks FROM Students GROUP BY Branch HAVING AVG(Marks) > 80;

Branch	Average_Marks
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CSE	87.4
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ECE	82.0
-----	------

EEE	81.0
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Task 4 — SQL Joins — 15 Marks

Demonstrated INNER, LEFT and equivalent RIGHT-join result coverage using related student/department tables.

Task 4 — SQL Joins

SQL: SELECT s.Student_ID, s.Name, d.Branch_Name FROM Students s INNER JOIN Departments d ON s.Branch = d.Branch_Name;

Student_ID	Name	Branch_Name
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101	Sai	CSE
102	Rahul	ECE
103	Priya	CSE
104	Arjun	IT
105	Anjali	CSE
106	Kiran	EEE
107	Sneha	CSE
108	Ravi	IT
... 2 more row(s)		

SQL: SELECT s.Student_ID, s.Name, d.Branch_Name FROM Students s LEFT JOIN Departments d ON s.Branch = d.Branch_Name;

Student_ID	Name	Branch_Name
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101	Sai	CSE
102	Rahul	ECE
103	Priya	CSE
104	Arjun	IT
105	Anjali	CSE
106	Kiran	EEE
107	Sneha	CSE
108	Ravi	IT
... 2 more row(s)		

SQL: SELECT d.Branch_ID, d.Branch_Name, s.Student_ID, s.Name FROM Departments d LEFT JOIN Students s ON s.Branch = d.Branch_Name;

Branch_ID	Branch_Name	Student_ID	Name
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1	CSE	105	Anjali
1	CSE	103	Priya
1	CSE	101	Sai
1	CSE	107	Sneha

Task 5 — SQL Subqueries — 10 Marks

Used subqueries for above-average salary and highest-salary problems.

Task 5 — SQL Subqueries

```
SQL: SELECT Employee_ID, Name, Salary FROM Employees WHERE Salary > (SELECT AVG(Salary) FROM Employees);
```

Employee_ID	Name	Salary
203	Rahul	75000
205	Arjun	80000
207	Kiran	65000
208	Meena	70000

```
SQL: SELECT Employee_ID, Name, Salary FROM Employees WHERE Salary = (SELECT MAX(Salary) FROM Employees);
```

Employee_ID	Name	Salary
205	Arjun	80000

Task 6 — Excel Formatting, Sorting & Filtering — 10 Marks

Prepared and formatted a student dataset, with sorting/filter evidence.

Student_ID	Name	Branch	Marks	Attendance	City
101	Sai	CSE	85	92	Hyderabad
102	Rahul	ECE	78	85	Vijayawada
103	Priya	CSE	92	95	Hyderabad
104	Arjun	IT	88	90	Warangal
105	Anjali	CSE	75	80	Hyderabad
106	Kiran	EEE	81	88	Karimnagar
107	Sneha	CSE	90	94	Hyderabad
108	Ravi	IT	72	76	Warangal
109	Meena	ECE	86	89	Vijayawada
110	Vijay	CSE	95	96	Hyderabad

Task 7 — Conditional Formatting & Excel Functions — 15 Marks

Demonstrated conditional formatting, IF, COUNTIF and SUMIF.

Student_ID	Name	Branch	Marks	Attendance	Result
101	Sai	CSE	85	92	Pass
102	Rahul	ECE	78	85	Fail
103	Priya	CSE	92	95	Pass
104	Arjun	IT	88	90	Pass
105	Anjali	CSE	75	80	Fail
106	Kiran	EEE	81	88	Pass
107	Sneha	CSE	90	94	Pass
108	Ravi	IT	72	76	Fail
109	Meena	ECE	86	89	Pass
110	Vijay	CSE	95	96	Pass
Number of CSE :		5			
Total Marks of C		437			

Task 8 — VLOOKUP & XLOOKUP — 10 Marks

Used related tables to retrieve branch names with VLOOKUP and XLOOKUP.

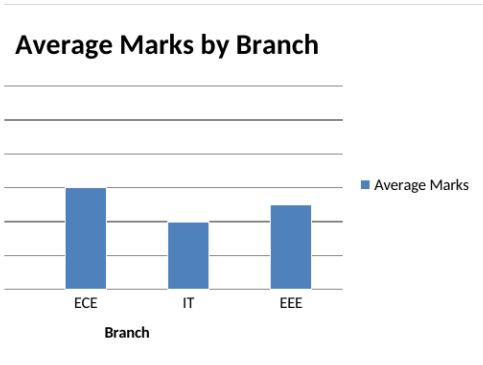
Student_ID	Name	Branch_ID
101 Sai		1
102 Rahul		2
103 Priya		1
104 Arjun		3
105 Anjali		1
106 Kiran		4
107 Sneha		1
108 Ravi		3
109 Meena		2
110 Vijay		1

VLOOKUP vs XLOOKUP

VLOOKUP	Searches vertically using a table range and column number.
XLOOKUP	Uses separate lookup and return ranges and is more flexible.

Task 9 — Pivot Table & Charts — 5 Marks

Created a branch summary and an average-marks chart.



Submission Checklist

- PDF report with task explanations and evidence images
- SQL files for Tasks 1–5
- Excel workbooks for Tasks 6–9
- Google Drive link with required files and correct access permission
- Week 3 GitHub repository link

Important: Before submitting, quickly open the report and verify that the evidence images match your final files. For Task 9, the workbook provided earlier contains the pivot-style summary and chart.